

Y9 5.2-5.3 bestest

Q1

One hundred cards are numbered from 1 to 100. A card is chosen at random. What is the probability that the first digit on the chosen card is 9?

$\frac{11}{100}$

Q2

Ten cards are numbered from 1 to 10. A card is chosen at random. What is the probability that the number on the chosen card is neither even nor a factor of 15.

$\frac{1}{5}$

Q3

A survey of 100 children in a village found that 95 of them had blue eyes. If this is representative of the 3600 people in the whole village, how many blue eyed people could you expect there to be in the village?

3420

Q4

A card is picked at random from a standard deck of 52 cards. Find the probability that it is the ace of hearts.

$\frac{1}{52}$

Q5

A box has 24 marbles that are either green or yellow. Lisa takes one marble out, notes the colour, then puts it back in the box. She does this 40 times.

Colour	Freq.
green	10
yellow	30

What is the relative frequency of green?

0.25

Using these results, what is the probability a marble drawn at random will be yellow?

% 75%

How many green marbles would you expect there to be in the box?

6

Q6

Musical instruments played.

	guitar (G)	not guitar (G̃)	
saxophone (S)	0.41	0.18	0.59
not saxophone (S̃)	0.15	0.26	0.41
	0.56	0.44	1

What percentage play the saxophone but not the guitar?

% 18%

What percentage play either the saxophone or guitar?

% 74%

Find the probability a student chosen at random plays both.

% 41%

Q7

A box contains one lemon, one raspberry and one cola ice-block. Jeremy chooses two ice-blocks at random.

Draw a tree diagram to help answer the question.

What is the probability the first ice-block is cola?

$\frac{\boxed{}}{\boxed{}} = \frac{1}{3}$

What is the probability he chose a raspberry followed by a lemon?

$\frac{\boxed{}}{\boxed{}} = \frac{1}{6}$

What is the probability he chooses a raspberry ice-block?

$\frac{\boxed{}}{\boxed{}} = \frac{2}{3}$

Q8

In a class of 28 students; 15 study Music, 21 study Visual Arts and 3 study neither.

What is the probability a student chosen at random studies both Music and Visual Arts?

$\frac{\boxed{}}{\boxed{}} = \frac{11}{28}$

What is the probability a student chosen at random studies Visual Arts but not Music?

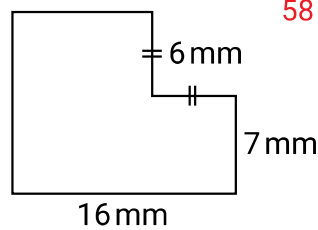
$\frac{\boxed{}}{\boxed{}} = \frac{5}{14}$

What is the probability two students chosen at random both study Music?

$\frac{\boxed{}}{\boxed{}} = \frac{5}{18}$

Q9

Find the perimeter. $\boxed{}$ mm

**Q10**

How many cubic centimetres in a cubic metre?

$\boxed{}$ 1 000 000

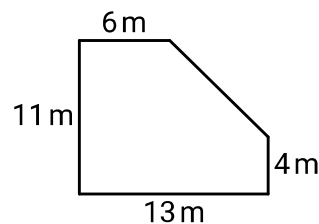
Q11

Find the surface area of a rectangular prism with length 5 cm, width 4 cm and height 7 cm.

$\boxed{} \text{ cm}^2$ 166

Q12

Find the area.

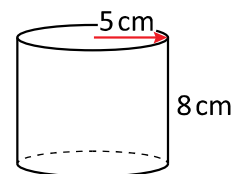


Area = $\boxed{} \text{ m}^2$ 118.5 m²

Q13

Find the surface area of this cylinder.

$\boxed{} \text{ cm}^2$ (1 d.p.) 408.4



Q14

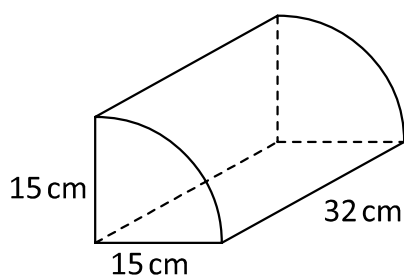
A rectangular room measures 3300 mm by 1250 mm and is 2.4 m from floor to ceiling. Calculate the volume of air the room contains in cubic metres.

m³ **9.9 m³**

Q15

A cylindrical tub of ice cream holds 1 litre. If the tub is 10 cm tall, find its radius. Answer in centimetres correct to 1 decimal place.

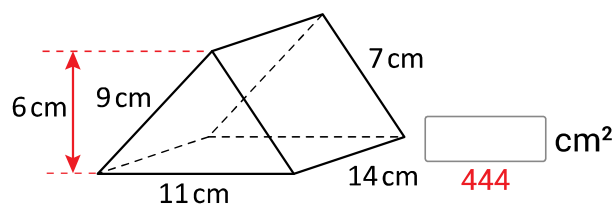
cm **5.6 cm**

Q16

Volume = cm³ (1 d.p.)
5654.9

Q17

Find the surface area of the triangular prism.



cm²
444

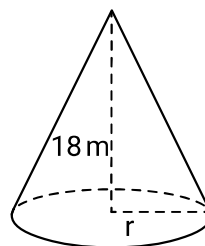
Q18

The volume of a pyramid is 500 m³. If the area of the base is 180 m², what is the height of the pyramid?

$h =$ m (1 d.p.)
8.3

Q19

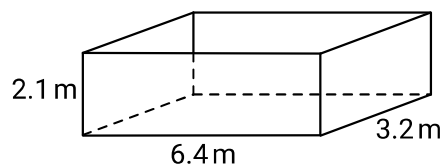
The volume of the cone is $600\pi\text{m}^3$ and its height is 18 m. Find the radius.



radius = m
10

Q20

A shipping container converted into a pool is shown.



Find the surface area of the inside of the pool.

m² (1 d.p.) **60.8**

Find the cost of tiling the pool at \$85 per square metre.

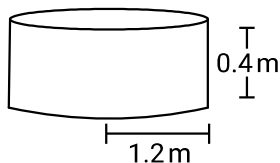
\$ nearest ten dollars **5170**

Find the capacity of the pool.

L (nearest 10 litres) **43010**

Q21

The base and sides of this garden pond are made of metal. It is open at the top.



The area of metal used to make the pond is closest to

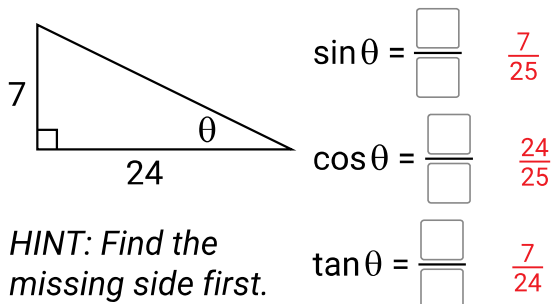
- ☒ 7.5m² ☐ 6.2m² ☐ 9.7m²

Find the volume of the pond.

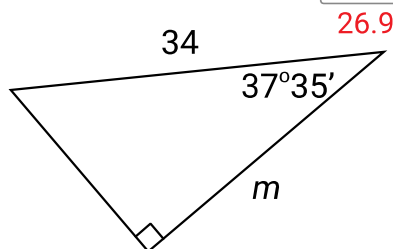
m³ (1 d.p.) 1.8

The capacity of the pond is closest to

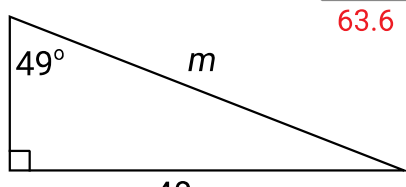
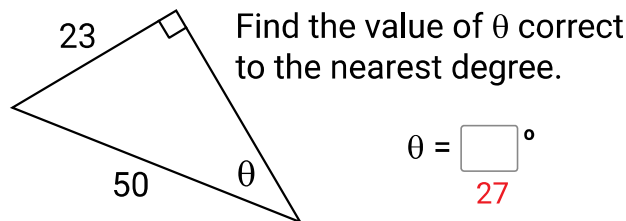
- ☐ 1785L ☒ 1810L ☐ 1825L

Q22**Q23**

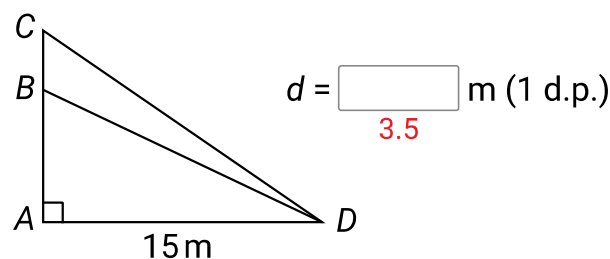
Find the value of m . (1 d.p.)

**Q24**

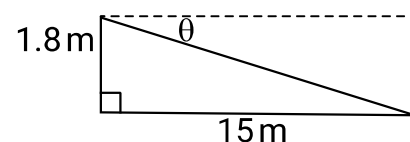
Find the value of m . (1 d.p.)

**Q25****Q26**

The angle of elevation from D to B is 25° .
The angle of elevation from D to C is 35° .
 AD is 15m. Find the distance from B to C .

**Q27**

A shark is swimming 15m from a jetty which is 1.8m above the water. Find the angle of depression to the shark.

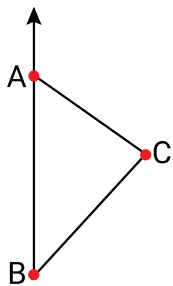


$\theta = \text{input}^\circ$ (nearest degree)

7

Q28

A is due north of B. C bears 125° from A and 042° from B.



Find the bearing of B from A.

$^\circ$ 180

Find the bearing of A from C.

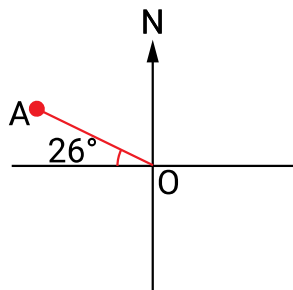
$^\circ$ 305

Find the bearing of B from C.

$^\circ$ 222

Q29

Write the 3-figure bearing of A from O.



Bearing = $^\circ$
296

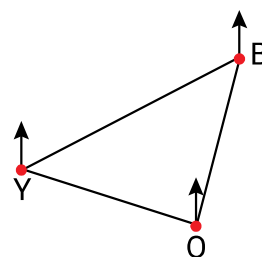
Q30

Max and Luke are standing on opposite sides of a 15m tall flagpole. The angle of elevation from Max to the top of the pole is 23° and from Luke to the top of the pole is 17° . How far apart are Max and Luke?

$d =$ m (1 d.p.)
84.4

Q31

O bears 110° from Y. B bears 065° and 023° from Y and O respectively.



$\angle BYO =$ $^\circ$ 45

$\angle BOY =$ $^\circ$ 93

$\angle OBY =$ $^\circ$ 42